APSIM NEXTGEN · CUSTOM CROP MODEL

A Day-Neutral Strawberry Model for Repeat-Flowering Production Systems

Development, calibration and field validation summary — phenology and biomass partitioning for Albion and San Andreas, calibrated against a controlled greenhouse trial and validated against two independent field seasons at Nambour, Queensland.

Prepared for the project lead

19 July 2026

01

Executive summary

Over this project we built a custom strawberry crop model inside APSIM NextGen, covering the full cycle from transplant through repeat flowering to fruit maturity, and tested it against real trial data from two very different growing environments: a controlled short-day greenhouse trial in Alabama, and two full field seasons at Nambour, Queensland, driven by observed SILO weather.

The phenology model — one shared parameter set, never retuned per site or cultivar — predicts days to first maturity within about 4 days of observed on average across all eight site-cultivar-season combinations tested (R^2 = 0.93). Fruit yield validation is still preliminary, resting on only the two Nambour field seasons where yield was actually measured; that comparison is reported honestly below rather than glossed over.

Alongside the model itself, we built a validation pipeline (mirroring APSIM's own Sorghum example) that automatically merges observed and simulated results and produces the comparison graphs in this report directly from the simulation database — so this report regenerates from the live model, not from a one-off analysis. We also produced a daily-resolution growth animation of the 2024 Nambour season, delivered both as an interactive web page and as a standalone video file.

Headline numbers

8	0.93	+0.6 d
simulated site-seasons validated — 2 field, 6 greenhouse	R^2, days to maturity — n=8, RMSE 3.7 d	bias, days to maturity — essentially unbiased

What's delivered

- Strawberry.apsimx — the runnable model: 8 simulations spanning 3 planting cohorts × 2 cultivars plus 2 independent field seasons.
- An automated observed-vs-predicted validation pipeline, with combined scatter graphs and per-simulation time-series panels.
- A daily-resolution animation of the AusQld_Nambour_2024 season (phenology stage + biomass partitioning), as an interactive page and a shareable video file.

02

Model overview

The model represents a day-neutral strawberry plant carried through repeated flowering cycles from a single transplant, rather than the single-flush annual habit APSIM's crop templates assume by default. Each cycle runs from flower induction through to harvest, then loops back to induce the next flush — the mechanism that lets one autumn planting at Nambour carry through to five fruiting cycles by the following spring.

Phenological sequence

Seven phases carry the plant from transplant to harvest; four of them (shaded) repeat for every subsequent flowering flush within a season.

Phase	Thermal-time target	Repeats each cycle
Transplanting to Established	60 °Cd	no
Established to Vegetative growth	500 °Cd	no
Flower induction to Anthesis	195 °Cd	yes
Anthesis to Fruit set	45 °Cd	yes
Fruit set to Green fruit	80 °Cd	yes
Green fruit to Maturity	120 °Cd	yes
Maturity to Harvest	30 °Cd	yes

Thermal time itself uses a Wang & Engel beta response rather than a simple linear above-base-temperature model: development rate rises from a base temperature, peaks at an optimum, and falls back to zero at a maximum, evaluated on a 3-hourly interpolated temperature curve rather than daily max/min. Cardinal temperatures are 5 / 20 / 30 °C (base / optimum / maximum) — see the parameter table below. Dormancy and chilling requirements were explicitly scoped out: the target production systems are grown as an annual crop from spring-established transplants, not carried over winter.

Model dynamics: phenology → leaf area → growth → partitioning

The cascade runs in one direction each day. Phenology (above) sets how fast the plant moves through its phase sequence, including how many times it loops back through flowering. Daily biomass supply is radiation-driven: intercepted photosynthetically active radiation — via Beer's law and a fixed canopy extinction coefficient — is converted to assimilate at a fixed radiation use efficiency (RUE). Leaf area itself is not tracked through a separate leaf-appearance or leaf-expansion sub-model; it is computed directly from live leaf biomass via a specific leaf area (SLA) constant, so canopy expansion is a straightforward biomass-times-SLA calculation rather than an explicit leaf-count/leaf-size process. This is a real simplification worth naming plainly, not a hidden one.

That daily assimilate is partitioned to Leaf, Crown and Root by fixed phase-dependent fractions — one split for the vegetative phase, a different split once fruiting begins at Fruit Set — rather than a dynamic priority scheme. Fruit is the exception: instead of a fixed fraction, it demands dry matter toward a target harvest-index trajectory once fruiting starts, and the Arbitrator's relative-allocation method reconciles that demand against the other organs' fixed-fraction demands for whatever supply is available that day. Crown additionally carries a small carbohydrate reserve pool (adapted from APSIM's Grapevine template) that fills during growth and can retranslocate into fruit filling when daily supply is short.

The one scope limit worth flagging directly to an agronomist: only development rate (phenology) responds to temperature. The RUE model's own temperature, water-stress and nitrogen-stress multipliers are all fixed at 1.0 in this version — daily biomass production is not yet penalised by heat, cold, drought or nitrogen limitation, only phenological timing is. This is deliberate "potential growth" scope for this phase of the work, not an oversight, but it means the model will not yet reproduce a season with a real water or nitrogen constraint.

Biomass partitioning

Daily assimilate is partitioned across four organs — leaf, crown, root and fruit — each tracked as its own dry-matter pool. Leaf area index and crown storage are tracked alongside these pools and feed back into the plant's light interception and the energy available for the next flowering cycle. This is the same partitioning the growth animation in Section 5 visualises day by day for a real simulated season.

Fresh vs dry weight

The model computes and partitions dry matter throughout; fresh fruit weight (used for the yield comparison in Section 4 and quoted anywhere as "g/plant") is derived from simulated dry weight by a single constant water-content assumption — 90% water, a literature-typical figure for strawberry fruit, not fitted from the McWhirt trial data (which did not report it) and not varying by cultivar, season, or ripeness stage. In other words, fresh weight is simply dry weight divided by 0.10. This is a reasonable starting assumption but a genuine simplification: any error in that 90% figure passes straight through into predicted fresh yield, and a measured dry-matter content for the actual fruit would tighten the yield validation in Section 4 more directly than almost any other single change.

Key input parameters

Parameter	Value	Status
Radiation use efficiency (RUE)	1.2 g DM/MJ	placeholder, not yet fitted
Light extinction coefficient (k)	0.58	Waister et al. 1980
Specific leaf area (SLA)	0.022 m^2/g live leaf DM	literature-typical, not fitted
Thermal time cardinal temps	5 / 20 / 30 °C	Wang & Engel beta, sub-daily
Leaf senescence rate	0.03 /d (~33 d lifespan)	placeholder, not fitted
Partition fractions, veg / fruiting	Leaf 45/20%, Crown 25/15%, Root 30/15%	placeholders
Crown reserve pool	max 30% of structural DM, fills 2%/d	adapted from Grapevine
Fruit harvest index (effective)	0.0095	rescaled - see note below
Fruit filling duration	200 °Cd	matches phenology targets
Fruit water content	90% (fixed)	literature-typical, not fitted

Parameter	Value	Status
Plant population	1 plant/m^2	explicit placeholder

The harvest index deserves a specific caveat. The literature value (0.35-0.50, Bringhurst & Voth 1984) is a per-flush harvest index; but with repeat flowering now looping the plant through Fruit Set to Maturity five to twelve times a season, applying that value on every pass overshoots real Nambour season yields by roughly 14-29 fold. The 0.0095 used here is a single shared value rescaled to sit between the two observed Nambour seasons, rather than a directly literature-sourced harvest index — it is a fitted stand-in, not a measured parameter, and is the most likely reason the two yield points in Section 4 bracket rather than match each season individually.

More broadly: the phenology thermal-time targets in the table above were calibrated against the McWhirt trial and are the best-supported numbers in the model. RUE, k, SLA, leaf senescence and the partition fractions are literature-typical placeholders that have not themselves been fitted to any of the observed data in this report — they happen to produce a good days-to-maturity fit because that fit is driven almost entirely by the phenology calibration, not by these growth parameters. They are the natural next target for calibration once more observed biomass or yield data is available (see Section 6).

03

Calibration & validation approach

Two independent data sources were used, deliberately kept separate so that validation results reflect genuine out-of-sample performance rather than a model re-fit to its own test data.

Calibration — McWhirt et al. 2023 (Auburn, AL, greenhouse)

Phase-duration targets were set against a controlled short-day greenhouse trial across three staggered planting cohorts (October, December and February) and both Albion and San Andreas cultivars — six simulations in total, used to fix the phenological parameters in Section 2.

Validation — Nambour, Queensland (2023 & 2024 field seasons)

Two full field seasons, each sown from a single April transplant and driven by observed SILO weather, were run with the calibration left untouched. These are the two simulations behind both the days-to-maturity and yield comparisons in Section 4, and the only genuinely independent test of the model against field-grown, not greenhouse, conditions.

Validation pipeline

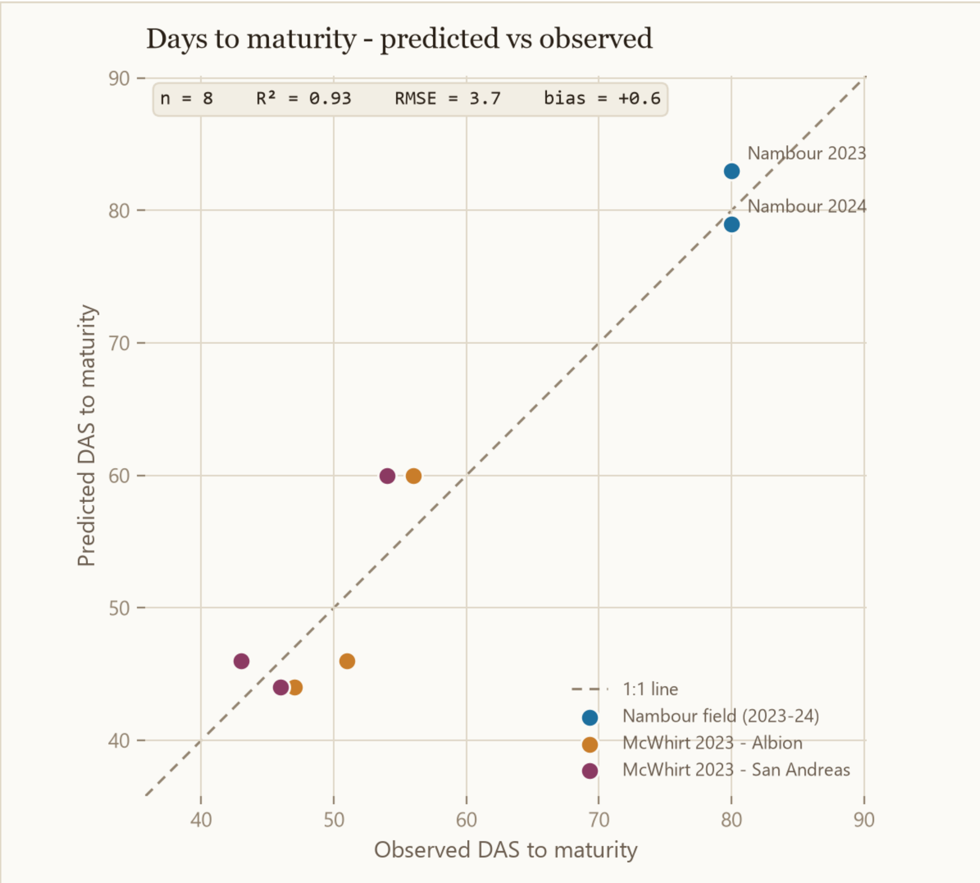
Observed records (first-harvest timing for all eight site-seasons, plus fresh yield for the two field seasons) were loaded into the model via an Excel input and merged against simulated output on matching day-after-sowing and simulation identifiers. This produces the combined observed-vs-predicted comparisons in Section 4, plus a per-simulation panel of phenology and biomass time series for each of the eight simulations — the same architecture APSIM's own Sorghum validation example uses, applied here to strawberry.

04

Combined results

Both comparisons below are generated directly from the model's own results database, so they reflect the current, live state of the calibration — not a point-in-time snapshot.

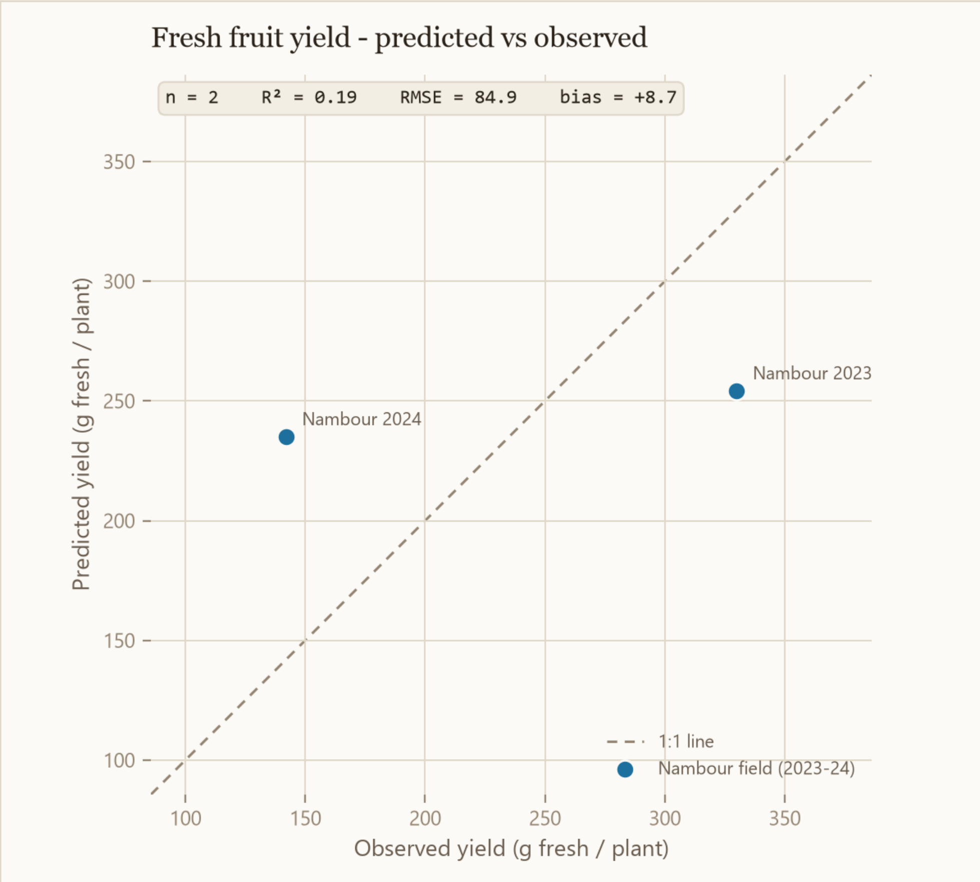
Days to maturity



Observed vs predicted days after sowing to first fruit maturity, across all 8 site-cultivar-season combinations. Colour distinguishes the calibration trial (McWhirt, by cultivar) from the two independent Nambour field seasons.

The fit is tight and essentially unbiased (R^2 = 0.93, RMSE = 3.7 days, mean bias +0.6 days) across both cultivars and both the greenhouse calibration trial and the two independent field seasons — evidence the phenology model transfers across the greenhouse-to-field gap without being re-tuned for it.

Fresh fruit yield



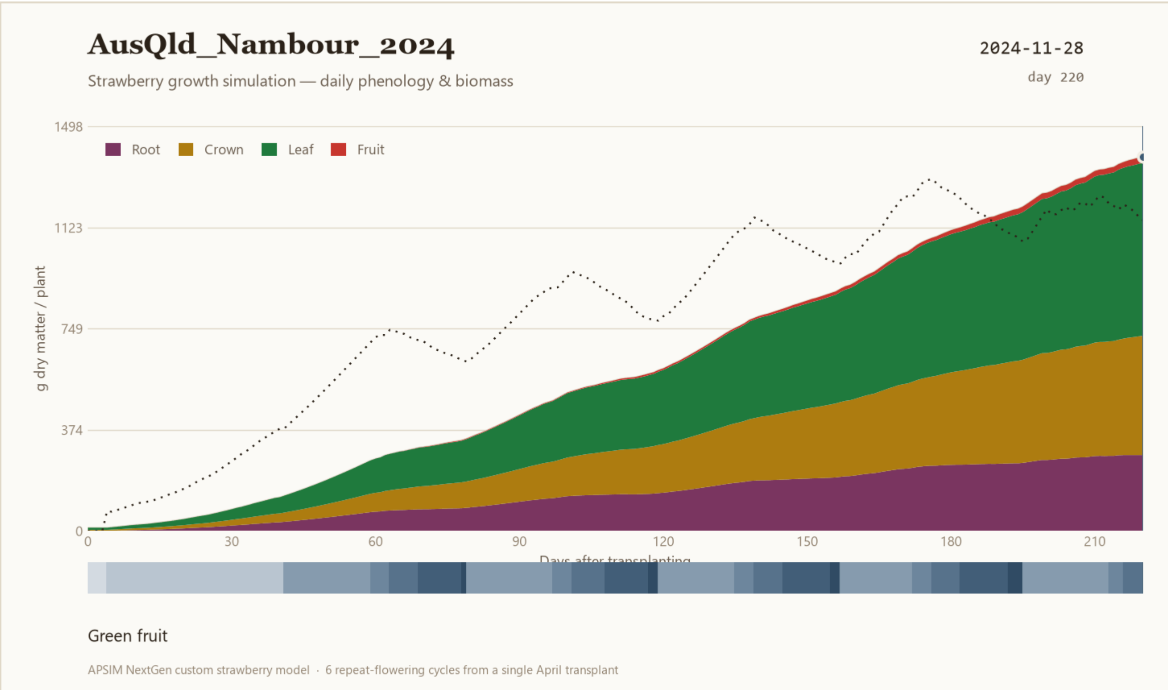
Observed vs predicted fresh yield per plant, the two Nambour field seasons — the only site-seasons where yield was recorded.

Yield validation rests on just two data points, so the statistics (R^2 = 0.19, RMSE ≈ 85 g/plant) should be read as a first look, not a settled result. The direction of error is not even consistent between the two seasons — 2023 is under-predicted, 2024 is over-predicted — which points to fruit carbon partitioning as the area most in need of more observed yield data and further refinement. Two contributing factors are named directly in Section 2: the fitted (not measured) harvest index that stands in for the repeat-flowering-compounded literature value, and the constant 90% fruit water-content assumption used to convert to fresh weight.

05

Growth animation — AusQld_Nambour_2024

To make the daily behaviour behind these summary statistics easier to inspect, we built a day-by-day animation of the 2024 Nambour simulation: 221 simulated days across five flowering cycles from the April transplant, showing the current phenological phase alongside the leaf / crown / root / fruit biomass split as it builds through the season.



Final frame of the animation (day 220, late in the fifth flowering cycle) — stacked biomass by organ, with the phenology phase strip below tracking all five repeat cycles.

Delivered in two formats

- Interactive page — scrubbable day-by-day, with play / pause and speed controls, light and dark themes.
- Nambour_2024_Growth_Animation.gif — a standalone video file for sharing outside the model, same visual design.

06

Deliverables & recommended next steps

Project files

File	Contents
Strawberry.apsimx	the runnable model — 8 simulations, validation pipeline
Observed.xlsx	observed trial data feeding the validation pipeline
build_apsimx.py	build script — regenerates the model file from source
Nambour_2024_Growth_Animation.gif	exported growth animation

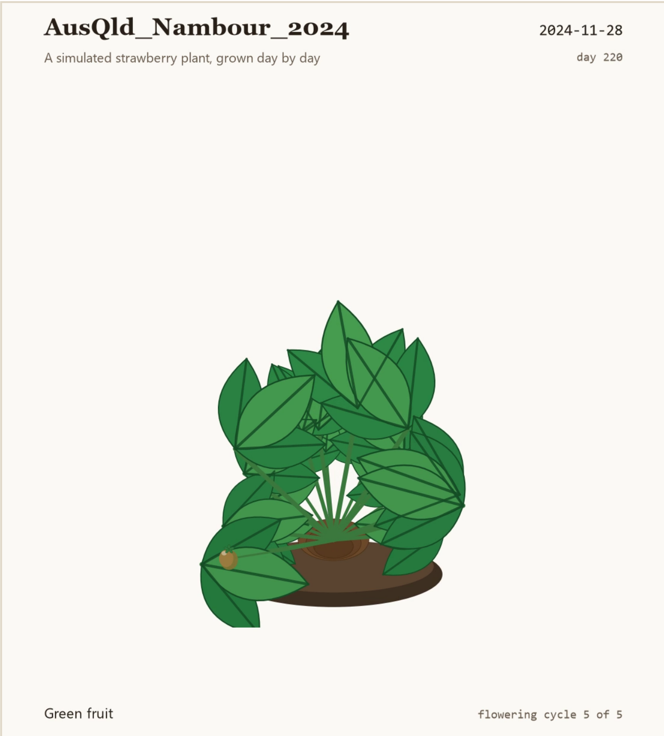
Recommended next steps

- Broaden the yield validation dataset — currently only 2 site-seasons have measured yield; more observed harvests, especially across cultivars, would let fruit partitioning be calibrated with the same confidence as phenology.
- Calibrate RUE, SLA, leaf senescence and the organ partition fractions against real data — these are currently literature-typical placeholders (see Section 2), not fitted parameters, and are the natural next target once more biomass or yield observations exist.
- Replace the fitted stand-in harvest index and the fixed 90% fruit water content with values grounded in real Nambour fruit measurements, to remove two of the most likely sources of yield-prediction error.
- Run a formal sensitivity analysis on the calibrated thermal-time targets to quantify how much of the remaining days-to-maturity error is parameter uncertainty versus weather-driven.
- If the model is extended beyond annual, spring-established systems, revisit the dormancy/chilling scope that was deliberately left out of this phase.
- Extend the per-simulation panel graphs and this report's charts to any new site-seasons as they're added — both regenerate directly from the model's results database.

07

Illustrated plant animation

Where Section 5 visualises the biomass numbers directly, this companion animation renders an actual illustrated strawberry plant — leaves, crown and a single fruiting truss — and grows it day by day from the same AusQld_Nambour_2024 simulation.



Final frame (day 220): five flowering cycles of growth, leaf count and size scaled to simulated leaf dry-matter, the fruiting truss's stage and colour driven by the plant's actual phenology phase on each simulated day.

What's driven by the model, and what's illustrative

- Leaf count and size scale directly with simulated leaf biomass (5 leaves at transplant, growing to 19 by season's end); crown size scales with simulated crown biomass.
- The fruiting truss's stage — flower, green, ripening, ripe — is read directly from the plant's simulated phenology phase for that day, across all 5 real flowering cycles.
- The model tracks one aggregate fruit pool rather than individual berries, so "one truss per cycle, picked between cycles" is a reasonable illustrative reading of that phenology, not a literal berry count.

Delivered in three formats

- Interactive page and Nambour_2024_Plant_Animation.gif — same as the Section 5 deliverables.
- Nambour_2024_Plant_Animation.mp4 — attached to this PDF (look for the attachment / paperclip icon in your PDF viewer's sidebar) and included alongside the project files.